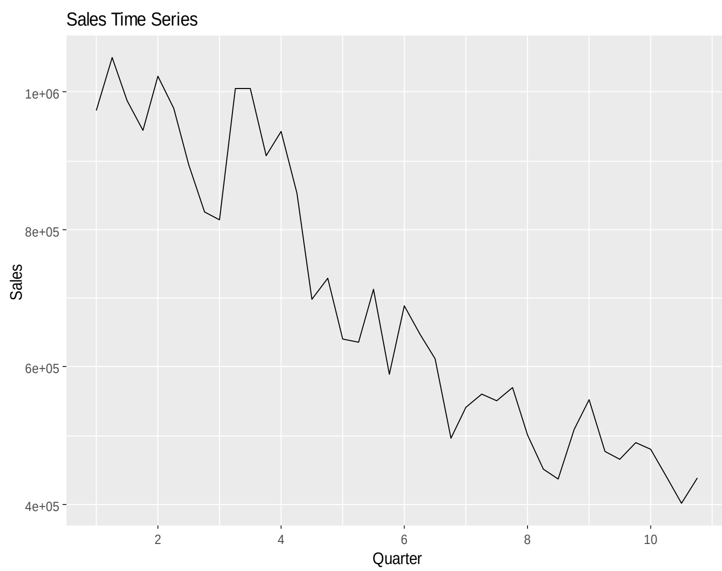
Tyson Test—Third Sales Dataset Analysis Document

For each series, I checked for heteroscedasticity by visually examining whether the variance appeared to change over time and then used a BoxCox.lambda() to determine whether a variance-stabilizing transformation was warranted.

I. Sales

1.1 Exploratory Data Analysis

Figure 1. Sales Time Series Plot



The Sales time series shows a clear overall downward trend over the sample period. Sales start relatively high, around a million, before subsequently declining over the time period before ending at around 400,000. While there are some short-run fluctuations and temporary rebounds, the dominant pattern remains sustained decline. The variation in the series also appears larger earlier in the sample than later, which suggests changing variance and supports checking whether a transformation may be necessary.

1.2 Seasonality

Figure 2. sales seasonal plot

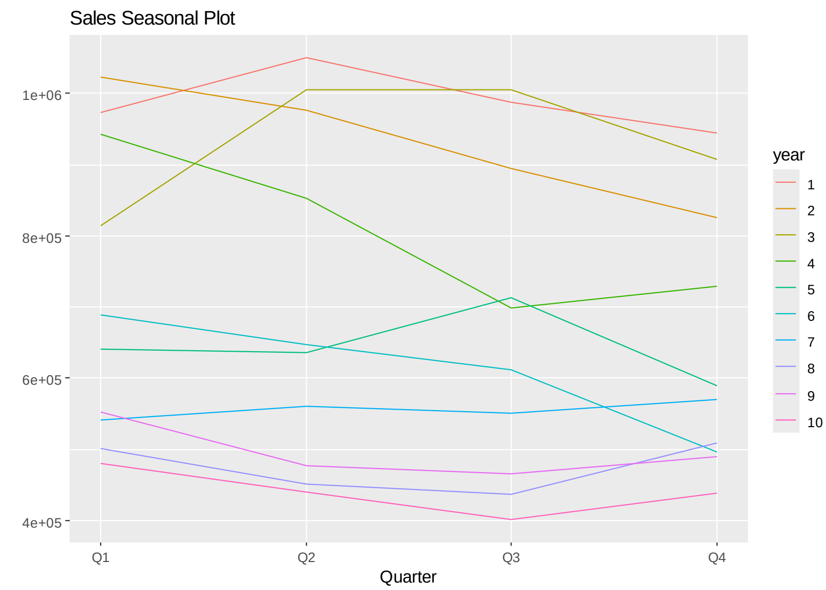
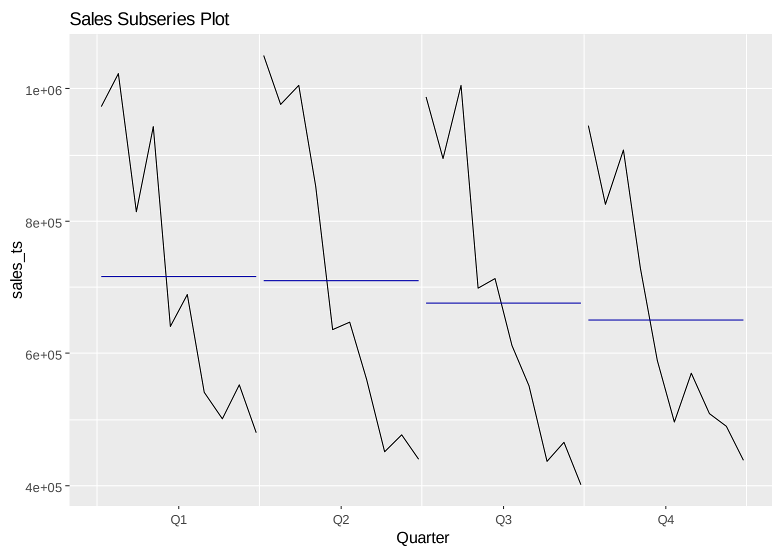


Figure 3. Sales subseries plot



The seasonal plots and subseries plot suggest that Sales do not follow an especially strong or stable seasonality across quarters. Although some years show quarter-to-quarter differences, the pattern is not highly consistent from one year to the next, and the dominant feature remains the broader downward movement in Sales over time. The subseries plot shows that average sales are somewhat higher in Q1 and Q2 than in Q3 or Q4, but the differences are not dramatic relative to the overall decline in sales over time. This suggests that seasonality may only be present weakly, which is consistent with checking nsdiffs() rather than relying on the plots alone.

1.3 Transformation

To check whether a transformation was needed, I estimated a Box-Cox lambda value for the Sales training series. The estimated lambda was about -0.08, which is not close to 1, indicating that a transformation is appropriate. Because the value was close to 0, the Box-Cox transformation is similar to a log-type transformation and is used to help stabilize the variance of the series. This was consistent with the visual impressions from the sales plot, where the variation appeared somewhat larger earlier in the sample than later.

1.4 Stationarity & Differencing

After transforming the Sales series, I tested whether the transformed data was stationary using the ADF and KPSS tests. The initial results suggested that the transformed series was still not stationary, since the ADF test did not provide enough evidence of stationarity at conventional levels and the KPSS test rejected stationarity. I then used ndiffs() to determine how many regular differences were needed, and the result indicated that one regular difference should be applied. After differencing once, the series appeared stationary: the ADF test supported stationarity, the KPSS test no longer rejected stationarity, and the Ljung-Box test from checkresiduals() produced a p-value of 0.6933, indicating no evidence of remaining autocorrelation. This suggests that transformation alone was not sufficient, but transformation combined with one regular difference was enough to stabilize the sales series before forecasting. Additional residual and autocorrelation diagnostics from checkresiduals() are reported in Appendix A.

1.5 Model Evaluation

To evaluate the competing forecasting methods for Sales, I compared Naïve, Seasonal Naïve, Random Walk with drift, ETS, and Auto.ARIMA using both the training-set and test-set accuracy measures. As per the rubric, I focused on the test-set MASE when selecting the best model, since MASE is scale-free and therefore more appropriate for comparing forecast performance than metrics based on the scale of the data. Among the five models, Random Walk with Drift produced the lowest test-set MASE, indicating that it produced the most accurate out-of-sample forecasts for the sales series. Because of this, it was selected as the preferred model.

Table 1. Sales training-set accuracy by model

Model	**ME**	**RMSE**	**MAE**	**MPE**	**MAPE**	**MASE**	**ACF1**
Naive	-14991.90	76703.59	62924.12	-2.6583	8.8720	0.5598	-0.1940
SNaive	-73530.14	134535.75	112397.79	-12.2770	16.9027	1.0000	0.4096
RWF_Drift	652.34	74512.16	59897.60	-0.5289	8.3799	0.5329	-0.1903
ETS	-18937.57	74384.90	60355.41	-3.3080	8.5250	0.5370	-0.0370
AutoARIMA	10880.57	67772.53	51127.35	0.3196	6.9854	0.4549	0.2569

Table 2. Sales test-set accuracy by model

Model	**ME**	**RMSE**	**MAE**	**MPE**	**MAPE**	**MASE**	**ACF1**	**Theil's U**
Naive	-41379.55	58673.98	52235.19	-9.6832	11.6470	0.4647	0.3126	1.6908
SNaive	-6341.30	37738.45	32975.27	-1.8370	7.0845	0.2934	0.4232	1.0231
RWF_Drift	2568.06	26773.95	21900.75	0.1138	4.6235	0.1949	-0.0530	0.5493
ETS	-25648.12	48869.16	40436.61	-6.2956	8.9709	0.3598	0.3126	1.3310
AutoARIMA	45511.14	53837.23	45511.14	9.4096	9.4096	0.4049	-0.1393	1.2523

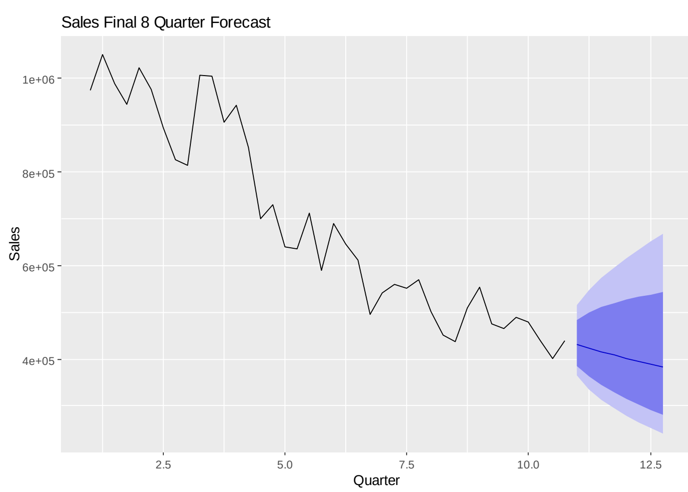
Table 3. Sales model comparison using test-set MASE

Model	**Test MASE**
Naive	0.4647
SNaive	0.2934
RWF_Drift	0.1949
ETS	0.3598
AutoARIMA	0.4049

1.6 Final Forecast

After selecting the random walk with drift, as the preferred model, I refit that model on the entire sales series and generated forecasts for the next eight quarters. The resulting forecast shows the continued negative trend over the forecast horizon, with predicted values declining from about 430,816 in the first forecasted quarter to 382,291 by the eight quarters ahead.

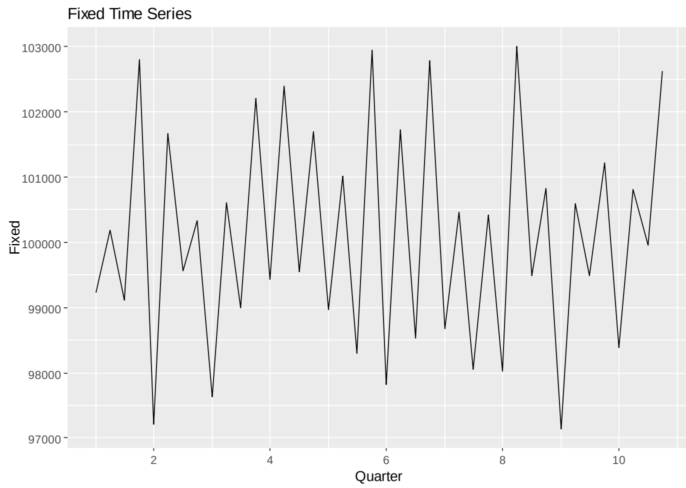
Figure 4. Sales final 8-quarter forecast



II. Fixed Costs

2.1 Exploratory Data Analysis

Figure 5. Fixed time series



The fixed series is much more stable than the Sales series and fluctuates over time within a relatively narrow range over the sample period. Most observations cluster near 100,000, with only modest quarter-to-quarter variation and no strong upward or downward trend. Compared with Sales, the Fixed series appears to be much less volatile, which is consistent with the idea that fixed costs tend to be fairly stable over time.

2.2 Seasonality

Figure 6. Fixed Seasonal Plot

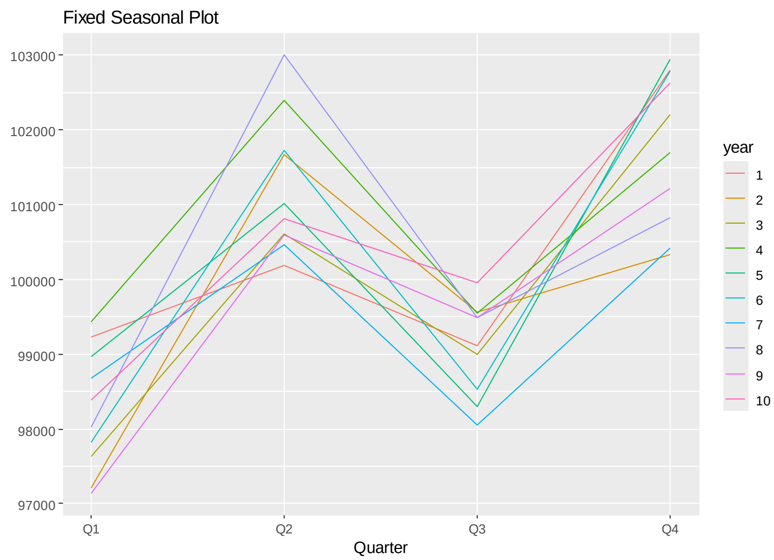
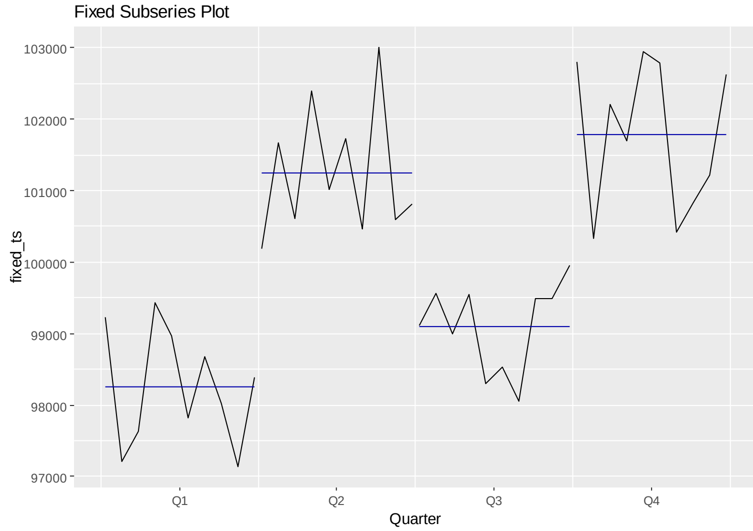


Figure 7. Fixed Subseries Plot



The seasonal and subseries plots indicate a fairly clear pattern in the fixed series, even though the overall variation is small relative to the other time series. In particular, q4 tends to have the highest fixed costs, while q1 has the lowest, q2 is also relatively high and q3 is somewhat lower. The subseries plot reinforces this by showing distinct average levels across the 4 quarters. This visual evidence is consistent with nsdiffs()=1, which suggests that seasonal differencing may be relevant. Overall, the fixed series looks pretty stable over time whilst also showing a noticeable and recurring pattern.

2.3 Transformation

To determine whether a transformation is needed for the Fixed series, I estimated a Box-Cox lambda using the training data. The estimated lambda was approximately -1, which is far from 1 and therefore indicates that a transformation should be applied. Because the value is close to -1, the transformation was more substantial than a simple log transformation and was used to help stabilize the variance of the Fixed series before stationarity testing and model fitting. Even though the fixed series appears relatively stable in level, the lambda result indicates that transforming the data was still appropriate.

2.4 Stationarity and Differencing

After transforming the fixed series, I tested whether the transformed data was stationary using the ADF and KPSS tests. The two gave somewhat mixed results: the ADF test did not return evidence of stationarity, while the KPSS test did not reject stationarity. I then used ndiffs() to determine whether regular differencing was needed, and it returned 0, indicating that no regular differencing should be applied. Even so, the Ljung-Box test from checkresiduals() produced a very small p-value, showing that strong autocorrelation remained in the transformed fixed series. This suggests that although regular differencing is not recommended, the series still contained a meaningful time dependence that needed to be handled by the forecasting models. Residual and autocorrelation diagnostics from checkresiduals() are reported in Appendix A and support the conclusion that strong autocorrelation remained in the transformed Fixed series.

2.5 Model Evaluation

To evaluate the competing forecasts for the fixed series, I compared naïve, seasonal naïve, random walk with drift, ets, and auto.arima using both the training set and test set accuracy measures. As in the sales section, I focused on the test-set MASE when choosing the best model because MASE is scale free and therefore more appropriate for comparing forecast performance than metrics that depend on the size of the data. Among the five candidates, ets produced the lowest test-set mase, making it the preferred model.

Table 4. Fixed training-set accuracy by model

Model	**ME**	**RMSE**	**MAE**	**MPE**	**MAPE**	**MASE**	**ACF1**
Naive	51.59	3196.02	2947.96	0.0007	2.9421	2.6143	-0.9079
SNaive	0.62	1323.98	1127.62	-0.0089	1.1239	1.0000	0.1111
RWF_Drift	-0.11	3197.26	2947.82	-0.0509	2.9427	2.6142	-0.9079
ETS	-12.24	837.61	737.74	-0.0190	0.7355	0.6542	0.0222
AutoARIMA	12417.70	35146.79	13403.83	12.3688	13.3600	11.8868	0.7343

Table 5. Fixed test-set accuracy by model

Model	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Naive	-795.94	1785.23	1346.87	-0.8216	1.3597	1.1944	-0.2969	0.5524
SNaive	-304.24	1374.84	1064.68	-0.3091	1.0583	0.9442	0.0107	0.6203
RWF_Drift	-1032.53	1842.50	1425.65	-1.0570	1.4406	1.2643	-0.3546	0.5805
ETS	-97.56	809.54	727.18	-0.1026	0.7274	0.6449	0.0983	0.3092
AutoARIMA	-304.24	1374.84	1064.68	-0.3091	1.0583	0.9442	0.0107	0.6203

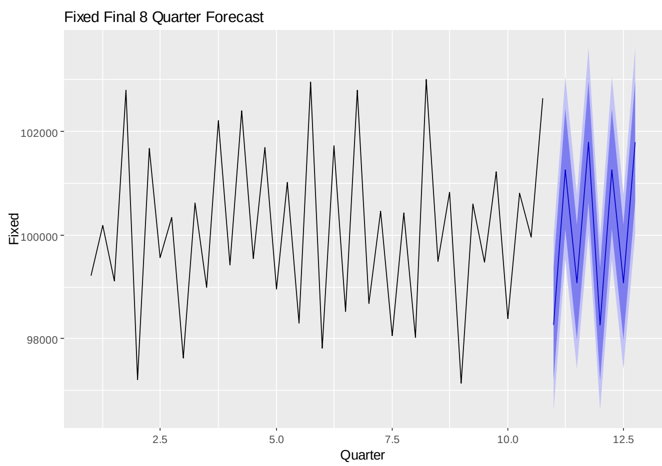
Table 6. Fixed model comparison using test-set MASE

Model	Test MASE
Naive	1.1944
SNaive	0.9442
RWF_Drift	1.2643
ETS	0.6449
AutoARIMA	0.9442

2.6 Final Forecast

After selecting ets as the best model for the fixed series, I refit the model using the entire series and generated forecasts for the next 8 quarters. The resulting forecast suggests that fixed costs will remain relatively stable over the forecast horizon and will continue to follow a repeating quarterly pattern rather than showing a strong upward or downward trend. Specifically, the forecast cycles through values of approximately 98,266.39, 101,257, 99,078, and 101,780, and then repeats this pattern over the second set of four quarters.

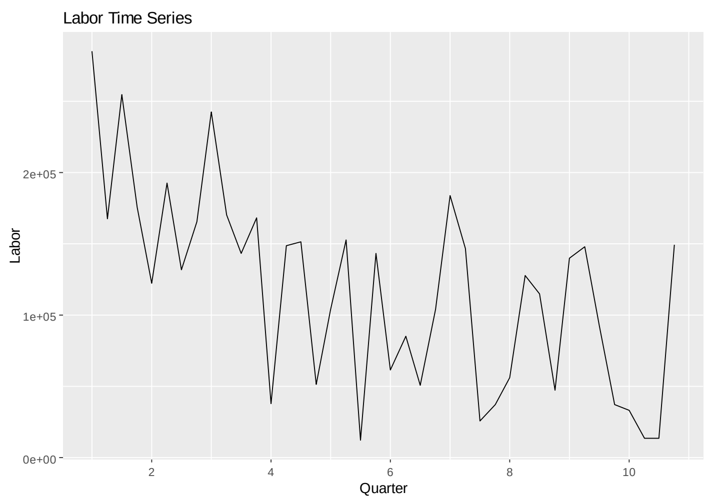
Figure 8. Fixed final 8 quarter forecast



III. Labor Costs

3.1 Exploratory Data Analysis

Figure 9. Labor Time Series



The labor time series is highly volatile and shows much more quarter-to-quarter fluctuations than either sales or fixed. The series starts at a relatively high level, but then moves irregularly, with several large rises and falls throughout the sample. There is no smoothing of long-run trend; instead, the dominant pattern is substantial short-run variation, including a few very low observations later in the series.

3.2 Seasonality

Figure 10. Labor Season Plot

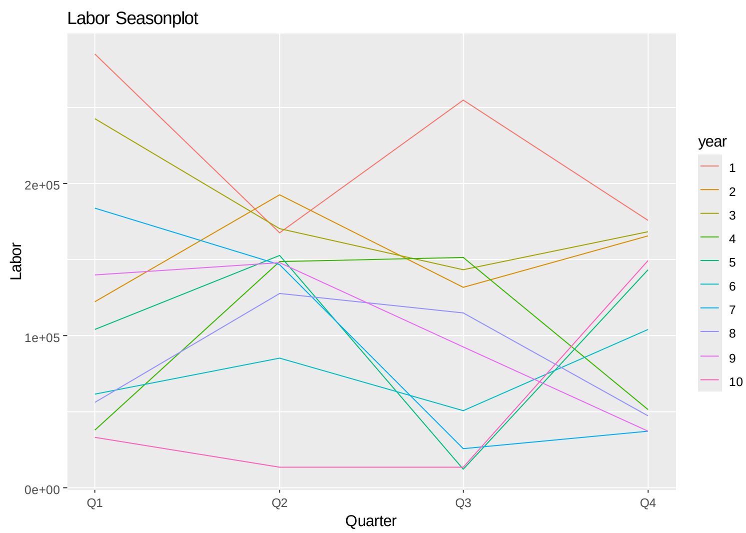
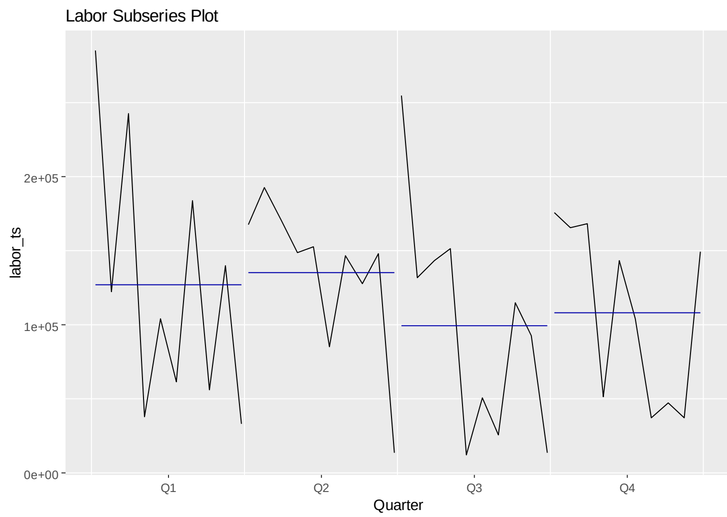


Figure 11. Labor Subseries Plot



The seasonal and subseries plots do not show a strong or consistent seasonal pattern in the labor series. The quarter-to-quarter pattern changes considerably across years rather than repeating in a stable way. The subseries plot also shows that average levels by quarter are not dramatically different from one another. This visual evidence is consistent with the nsdiffs() result of 0, which indicates that seasonal differencing is not needed for the labor series.

3.3 Transformation

To figure out if I needed to do a transformation for the labor costs series, I estimated a BoxCox lambda using the training data. The estimated lambda was approximately 1.12, which is close enough to 1 so that I didn't need to apply a transformation. So, I used the original labor series rather than transforming it before testing for stationarity and fitting the forecasting models.

3.4 Stationarity & Differencing

I tested the original labor training series for stationarity using the ADF and KPSS tests. The ADF result provided only borderline evidence of stationarity, while the KPSS test rejected stationarity. I then used ndiffs() to determine the amount of regular differencing needed, and it returned 1, showing that it should be differenced once. After first differencing, the stationarity results improved: the

ADF test supported stationarity, the KPSS test no longer rejected stationarity, and the differenced series could be treated as reasonably stationary. Because the series still had pattern left in it (as measured by the Ljung-Box test), it was worth comparing multiple forecasting methods to see which one handled the pattern best. Residual and autocorrelation diagnostics from checkresiduals() are reported in Appendix A and show that some autocorrelation remained even after differencing.

3.5 Model Evaluation

To evaluate the competing forecasting methods for the labor series, I compared naïve, seasonal naïve, random walk with drift, ets, and auto.arima using both the training-set and test-set accuracy measures. As with earlier sections, I focused on test-set MASE because it is a scale-free measure and is therefore appropriate for comparing forecast accuracy across different models. Based on the reported values, naïve was selected as the best model because it had the minimum displayed test-set MASE. Although ets and auto.arima produced the same displayed MASE value to 4 decimal points, which.min() selected the first minimum, which was naïve.

Table 7. Labor training-set accuracy by model

Model	**ME**	**RMSE**	**MAE**	**MPE**	**MAPE**	**MASE**	**ACF1**
Naive	-7655.51	75632.62	65470.35	-68.2127	108.7778	0.9995	-0.4540
SNaive	-19123.25	85096.81	65500.70	-82.8813	117.7910	1.0000	-0.1440
RWF_Drift	0.00	75244.18	65717.30	-57.9357	105.4515	1.0033	-0.4540
ETS	-15012.51	58418.97	48004.78	-69.5021	88.1591	0.7329	-0.1625
AutoARIMA	-19477.50	58415.36	46839.05	-70.6281	87.6996	0.7151	-0.1620

Table 8. Labor test-set accuracy by model

Model	**ME**	**RMSE**	**MAE**	**MPE**	**MAPE**	**MASE**	**ACF1**	**Theil's U**
Naive	31285.67	65007.20	54002.21	-38.4293	101.3557	0.8245	0.3494	0.8021
SNaive	-8070.77	72700.97	59584.54	-194.2993	229.7315	0.9097	0.0823	1.1229
RWF_Drift	65735.47	84057.40	65735.47	87.2256	87.2256	1.0036	0.1736	1.2153
ETS	-5477.27	57246.34	54002.21	-146.0938	180.1859	0.8245	0.3494	0.7590
AutoARIMA	-4482.71	57159.76	54002.21	-143.1811	178.0533	0.8245	0.3494	0.7572

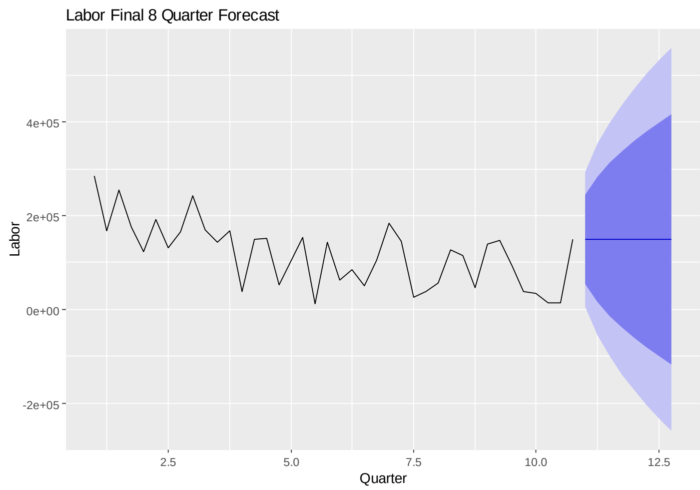
Table 9. Labor model comparison using test-set mase

Model	**Test MASE**
Naive	0.8245
SNaive	0.9097
RWF_Drift	1.0036
ETS	0.8245
AutoARIMA	0.8245

3.6 Final Forecast

After selecting the naïve model as preferred for labor, I refit it using the entire labor series and generated forecasts for the next eight quarters. The final forecast is flat, with each of the next eight quarters predicted to be approximately 149,456. This reflects the nature of the naïve model, which projects future values using the most recent observed value.

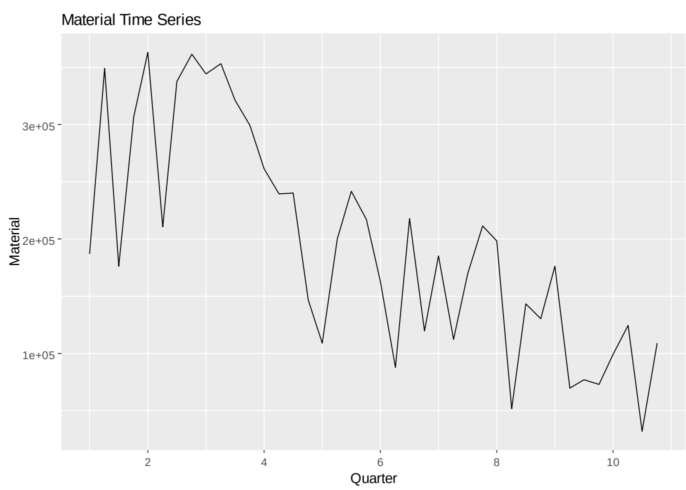
Figure 12. Labor final 8 quarter forecast



IV. Materials Costs

4.1 Exploratory Data Analysis

Figure 13. Material Time Series



The material series shows a clear downward trend over the sample period, falling from relatively high values at the start to much lower values later in the series. In addition to this downward trend, the series also shows short-run fluctuations, with sharp increases and decreases from one quarter to the next. Compared with fixed, the materials series is much more volatile, and unlike labor, the main visual pattern is a long-run decline rather than purely erratic movement. Overall, the material series looks like it combines trend with noteworthy variability.

4.2 Seasonality

Figure 14. Material Seasonal Plot

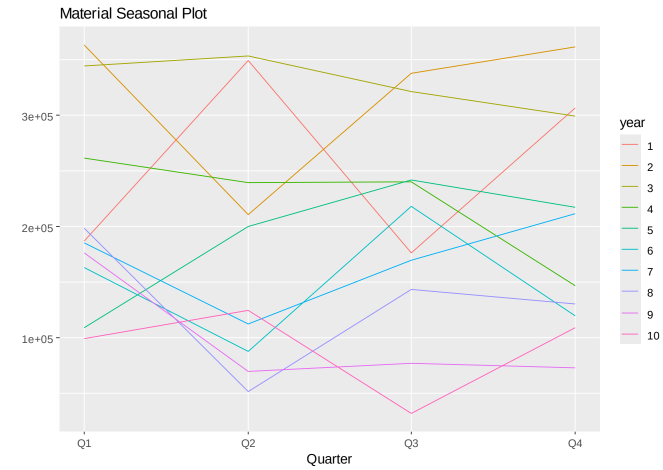
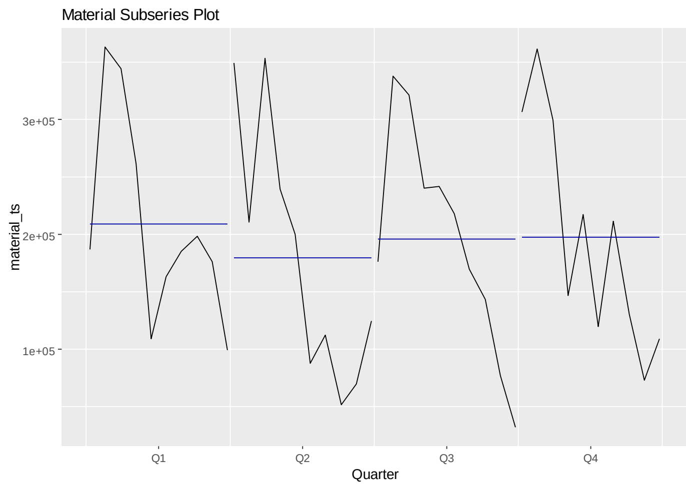


Figure 15. Material Subseries plot



The seasonal and subseries plots don't show particularly strong or stable seasonality in the material series. While some quarters are higher than others in particular years, the quarter-to-quarter pattern changes substantially across years rather than repeating it in a clear and consistent way. The subseries plot shows that q1 has the highest average material costs, and that the rest of the quarters are fairly close together, but these differences are modest compared to the overall downward trend and overall volatility of the series. This interpretation is consistent with nsdiffs()=0, which indicates that seasonal differencing is not needed.

4.3 Transformation

To determine whether a transformation was needed for the material series, I estimated a Box-Cox lambda of 0.76 using the training data. This is not close enough to 1 to justify leaving the data in its original scale. Therefore, I used a Box-Cox transformation before testing for stationarity and fitting the forecasting models. This choice was also consistent with the visual impression that the material series showed substantial variation and changing spread over time.

4.4 Stationarity & Differencing

After transforming the materials series, I tested whether the transformed data was stationary using the ADF and KPSS tests. The initial results suggested that the transformed data was not stationary, since the ADF test did not support stationarity and the KPSS test rejected stationarity. I then used ndiffs() to determine how many regular differences were needed. It returned 1, meaning that one regular difference should be applied. After first differencing, the stationarity results improved: the ADF test supported stationarity, the KPSS test no longer rejected stationarity, and the Ljung-Box test did not indicate significant remaining autocorrelation. This all means that this series would benefit from both a Box-Cox transformation and one regular difference before forecasting. Residual and autocorrelation diagnostics from checkresiduals() are reported in Appendix A and are consistent with the conclusion that differencing improved the behavior of the series.

4.5 Model Evaluation

To evaluate the competing forecasting methods for the materials series, I compared naïve, seasonal naïve, random walk with drift, ets, and auto.arima using both training set and test set accuracy measures. I again focused on the test-set mase to identify the best model, since it is a scale-free measure and therefore well suited for forecast comparison. Among the five candidates, random walk with drift produced the lowest test-set mase, showing that it gave the most accurate forecasts for the material cost series.

Table 10. Material training-set accuracy by model

Model	**ME**	**RMSE**	**MAE**	**MPE**	**MAPE**	**MASE**	**ACF1**
Naive	-1813.60	84561.21	68319.20	-15.0999	42.6835	0.9023	-0.4877
SNaive	-17701.05	91040.17	75720.00	-22.0961	43.6270	1.0000	0.0511
RWF_Drift	123.92	84443.13	68053.25	-14.0356	42.2920	0.8987	-0.4875
ETS	-9314.97	67973.52	58121.14	-19.2212	36.8380	0.7676	-0.0920
AutoARIMA	-5748.76	68053.51	56829.29	-17.4299	35.9133	0.7505	-0.0529

Table 11. Material test-set accuracy by model

Model	**ME**	**RMSE**	**MAE**	**MPE**	**MAPE**	**MASE**	**ACF1**	**Theil's U**
Naive	-35721.73	53909.38	47017.93	-71.8164	78.2371	0.6209	-0.2977	0.6126
SNaive	-35903.26	67540.10	58676.48	-69.7247	90.9347	0.7749	-0.5728	0.8657
RWF_Drift	-27990.75	48218.96	40670.27	-60.7058	68.1376	0.5371	-0.3340	0.4972
ETS	-39884.51	56753.45	50140.01	-77.2867	83.1159	0.6622	-0.2977	0.6700
AutoARIMA	-39083.86	56193.66	49539.52	-76.2346	82.1776	0.6542	-0.2977	0.6589

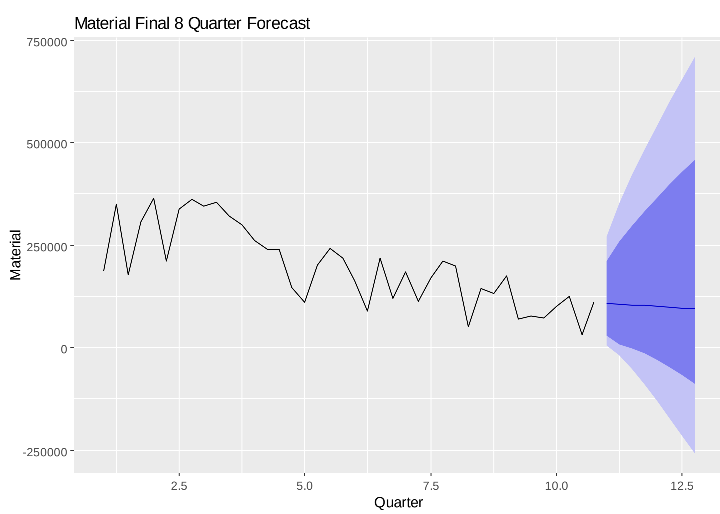
Table 12. Material model comparison using test-set mase

Model	Test MASE
Naive	0.6209
SNaive	0.7749
RWF_Drift	0.5371
ETS	0.6622
AutoARIMA	0.6542

4.6 Final Forecast

After selecting the random walk with drift as the preferred model for the materials series, I refit the entire series on that model and generated a forecast for the next 8 quarters. The final forecast indicates a continual decline in materials cost over the horizon, with predicted values falling from about 107,309 in the first quarter ahead to about 94,846 by the eighth quarter ahead. This suggests that the broader downward trend observed in the historical materials series is expected to continue over the next 2 years.

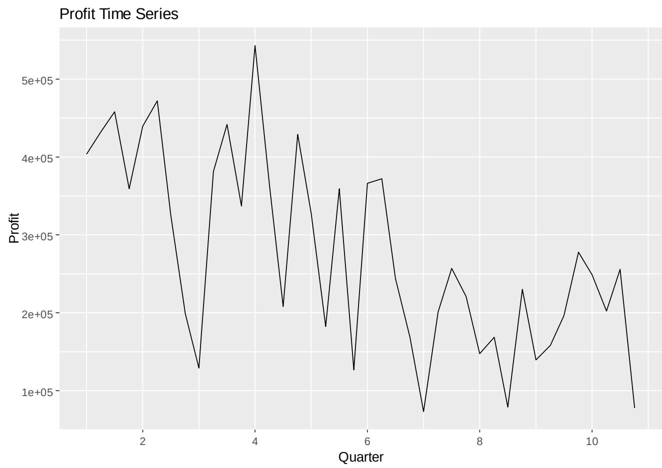
Figure 16. Material final 8 quarter forecast



V. Profit

5.1 Exploratory Data Analysis

Figure 17. Profit Time Series



The profit time series shows a clear overall downward trend over the sample period, although it also shows a lot of quarter-to-quarter volatility. The series starts at a relatively high level, with several early observations above 400k, but later observations are much lower and include several large dips. Compared with sales, the profit series appears much more irregular, with several large, short-run swings around a broader downward movement. Overall, the main visual pattern is declining profitability over time, combined with considerable volatility.

5.2 Seasonality

Figure 18. Profit seasonal plot

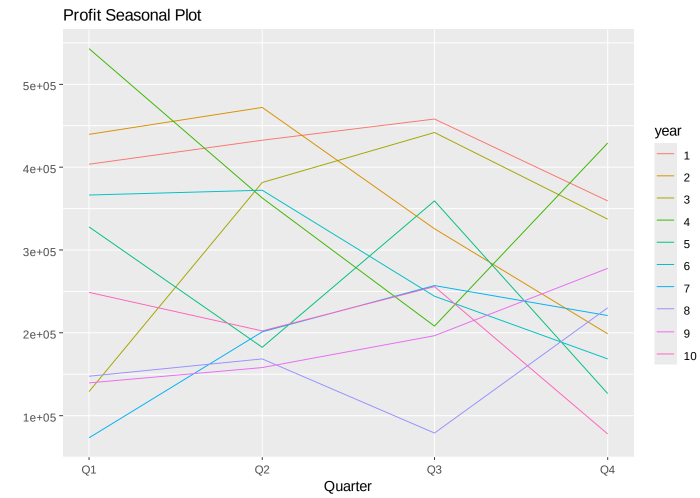
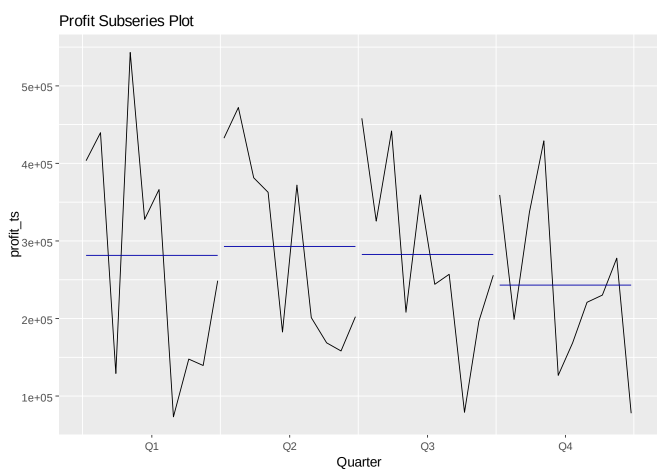


Figure 19. Profit subseries plot



The seasonal and subseries plots do not show strong or stable seasonality in the profit series. Though there are some quarter-to-quarter differences, the pattern changes noticeably across years rather than repeating in a consistent way. The subseries plot shows that q2 and q3 are slightly higher on average than q1 and q4, but these differences are modest relative to the broader downward trend and large fluctuations in the series. This interpretation is consistent with nsdiffs()=0, which indicates that seasonal differencing is not needed for profit.

5.3 Transformation

To determine whether or not a transformation was needed for the profit series, I estimated a BoxCox lambda using the training data. The estimated lambda was about 0.31, which is far enough from 1 to warrant a transformation. On that basis, I applied a Box-Cox transformation before testing for stationarity and fitting the forecasting models. This is also consistent with the visual appearance of the profit series, which showed substantial variation and changing spread over time.

5.4 Stationarity & Differencing

After transforming the profit series, I tested whether the transformed data was stationary using the ADF and KPSS tests. The initial results were mixed, with the ADF test suggesting stationarity but the KPSS test still rejecting it. I then employed ndiffs() to determine the number of regular differences needed, and it returned 1. This indicates that 1 regular difference should be applied. After differencing once, the profit series appeared stationary: the ADF test strongly supported stationarity, the KPSS test no longer rejected stationarity, and the Ljung-Box test was not significant at the 5% level. This suggests that a Box-Cox transformation together with one regular difference was sufficient to stabilize the profit series for forecasting. Residual and autocorrelation diagnostics from checkresiduals() are reported in Appendix A and support the conclusion that the differenced Profit series was acceptable for forecasting.

5.5 Model Evaluation

To evaluate the competing models for the profit series, I compared naïve, snaive, random walk with drift, ets, and auto.arima using both the training-set and test-set accuracy measures. As in the earlier sections, I used test-set mase for the same reasons as above. Among the five candidates, random walk with drift produced the lowest test-set mase, showing that it gave the most accurate out-of-sample forecasts for the profit series. Therefore, random walk with drift was chosen as the preferred model.

Table 13. Profit training-set accuracy by model

Model	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Naive	-5574.37	133212.58	114388.20	-18.0777	49.4836	0.8335	-0.3541
SNaive	-36706.47	171166.90	137240.36	-46.0979	74.2113	1.0000	0.0628
RWF_Drift	-313.17	132212.06	113701.66	-15.9115	48.5620	0.8285	-0.3542
ETS	-2937.92	100766.40	80127.78	-16.4068	37.9673	0.5839	0.1277
AutoARIMA	14688.72	87523.70	71684.32	-7.9240	31.4055	0.5223	0.0517

Table 14. Profit test-set accuracy by model

Model	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Naive	-36147.55	72592.59	58677.71	-38.3684	46.8662	0.4276	0.0052	0.9459
SNaive	37936.97	101502.80	80949.72	-1.0159	51.7465	0.5898	-0.3958	1.4922
RWF_Drift	-16494.16	66069.16	56498.31	-25.8431	41.1006	0.4117	0.0623	0.8687
ETS	77691.66	101333.80	82478.24	30.9237	37.1221	0.6010	0.0934	1.4764
AutoARIMA	55603.17	94341.42	84248.02	16.9291	42.8031	0.6139	-0.0033	1.3301

Table 15. Profit model-comparison using test-set mase

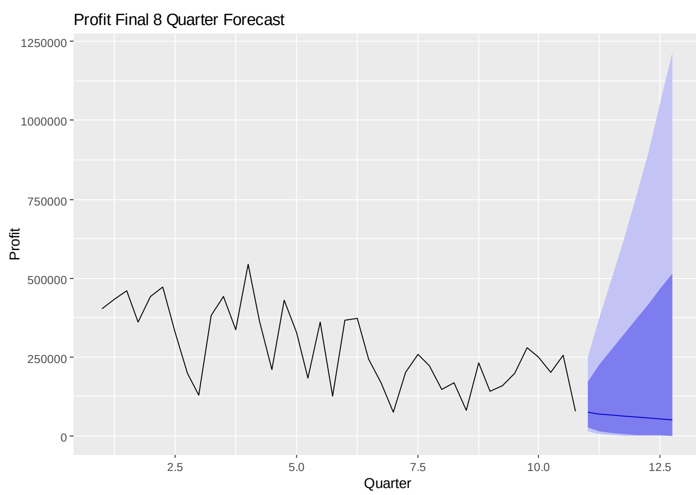
Model	Test MASE
Naive	0.4276
SNaive	0.5898
RWF_Drift	0.4117
ETS	0.6010
AutoARIM	0.6139

Model	Test MASE
A	

5.6 Final Forecast

After selecting the random walk with drift as the preferred model, I then refit the entire series on the model and generated forecasts for the next eight quarters. The final forecast indicates a continued decline in profitability over the forecast horizon, with predicted values falling from about 73,300 in the first quarter ahead to about 49,929 by the eighth quarter ahead. This suggests that the downward pattern observed historically will continue for the next 2 years.

Figure 20. Profit final 8 quarter forecast



VI. Final Forecast Table

Table 16 reports the final eight-quarter forecasts for sales, fixed, labor, materials, and profit using the preferred model for each series. These forecasts were made by refitting the chosen model to the full-time series and then projecting eight quarters into the future.

Table 16. Final eight-quarter forecasts for sales, fixed costs, labor costs, material costs, and profitability

Quarter Ahead	**Sales**	**Fixed**	**Labor**	**Material**	**Profit**
1	430815.93	98266.39	149456.00	107308.62	73300.01
2	423378.05	101257.40	149456.00	105499.10	69531.97
3	416117.38	99077.55	149456.00	103699.20	65913.89
4	409028.55	101780.27	149456.00	101908.98	62441.58
5	402106.38	98266.39	149456.00	100128.50	59110.91
6	395345.90	101257.40	149456.00	98357.82	55917.85
7	388742.29	99077.55	149456.00	96597.02	52858.40
8	382290.93	101780.27	149456.00	94846.16	49928.64

The combined forecast table shows that sales, material costs and profit are all expected to decline over the next eight quarters. Fixed costs follow a stable repeating pattern and labor costs remain flat.

VII. Conclusion

Overall, the forecasts suggest that this business is likely to struggle over the next two years. Sales, materials costs, and profit are all expected to decline, while fixed costs stay fairly steady and labor remains flat. Together, this all points to a worrying outlook for this firm, since revenue and profitability are both moving in the wrong direction while the costs are not declining fast enough to offset the revenue decline. If these trends don't abate soon, the long-term fiscal viability of this firm may be called into question.

Appendix A.

Figure A1. Sales checkresiduals() output

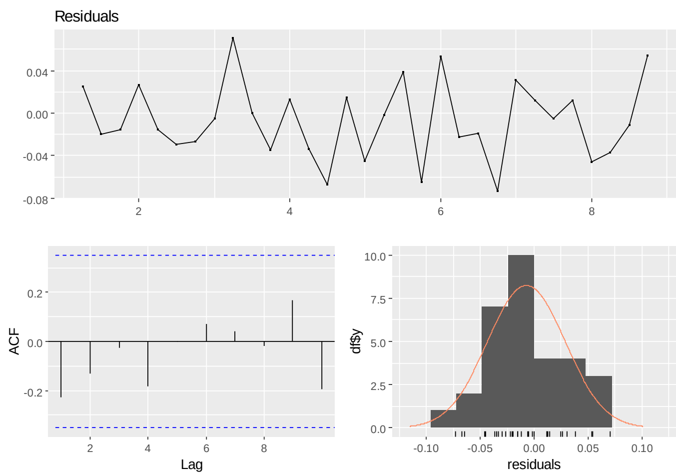


Figure A2. Fixed checkresiduals() output

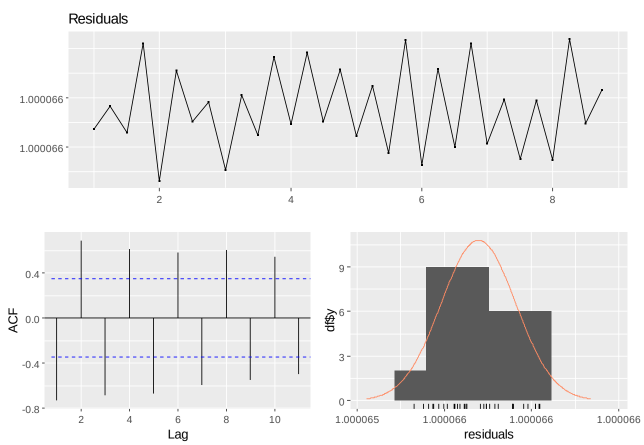


Figure A3. Labor checkresiduals() output

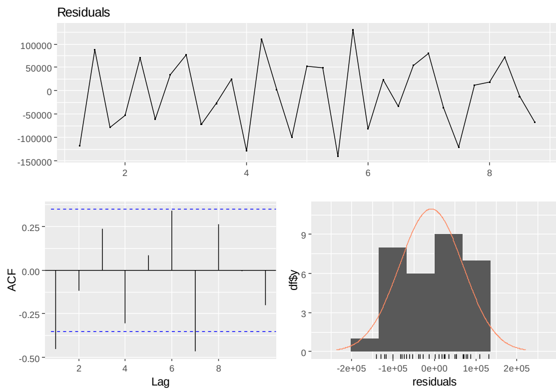


Figure A4. Material checkresiduals() output

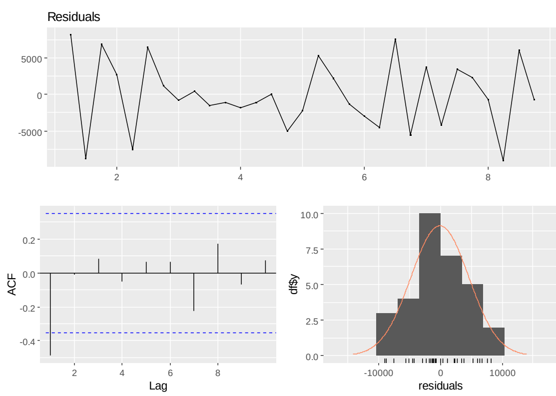


Figure A5. Profit checkresiduals() output